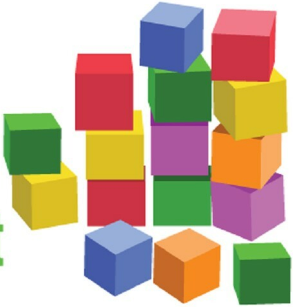
Starting Out with Maemo App Development



Maemo development can be divided into two broad categories — application development and platform development. In this article, we will not go into excessive details about developing the platform but focus more on getting started with application development on Maemo 5.

Maemo is derived from the Debian GNU/Linux distribution and as a package it combines the Maemo OS and the SDK. Maemo consists mainly of the software stack from the Linux kernel to the Maemo APIs and the Hildon UI framework. Using Maemo, you can develop your own applications atop the default Hildon UI provided by the supplier.

The UI architecture of Maemo is based on the GNOME desktop with many derivations from the GTK+ framework, GStreamer for multimedia, GConf for configuration and an XML library.

Setting up the development environment

All the application development for Maemo is done with the help of the Maemo 5 SDK. The SDK creates a Maemo development environment on your system. It's a sandbox environment built on a tool called Scratchbox—it behaves as a Maemo-based device on your system and has the tools to support the development. See the sidebox on Scratchbox for more information.

It is similar to a hardware emulator designed as a software simulator. I will not go into the details about installing Maemo on your Linux system. If you use Debian derivatives then the chances are that a binary package is already available in your repositories, which you should search for and install. I still urge you to read through the official installation guide on the Maemo Community website at *http://wiki.maemo.org/Documentation/Maemo5_Final_Installation*. Once you have finished with the installation of the SDK, you can begin tuning it for development.

For the emulator in the Scratchbox to work, you need the *binfmt_misc-feature* module loaded. Check whether or not it is loaded with *lsmod | grep binfmt*. If not, then as the root user, execute *modprobe binfmt_misc*. If it does not work, check the documentation for your Linux system. It may be directly embedded in the kernel instead of being a module. In the worst-case scenario, you will have to compile your own kernel from source.

Once you develop the application, you will need an X server to test it. For this, you need to install a psuedo X server. We will be focusing on Xephyr in this tutorial. Xephyr is an X server that can emulate 16-colour depth for its clients even while acting as a client to a 24-bit depth real X server. Xephyr also implements modern X protocol extensions. Install Xephyr from your distro's repositories.

As mentioned before, Maemo SDK uses Scratchbox as the cross-compilation environment. Therefore, you must install Scratchbox first before attempting to install